Investigating CTT in RNNs: Kill/Birth Ratios and Hebbian-based Scoring

Nathan Del Fava
University of Cape Town
Cape Town, South Africa
DLFNAT001@myuct.ac.za

Ziyaad Shaboodien
University of Cape Town
Cape Town, South Africa
SHBZIY001@myuct.ac.za

1 Introduction

A fundamental challenge in the design of artificial neural networks is the balance between stability and plasticity. The challenge manifests most commonly through catastrophic forgetting, whereby task performance for neural networks significantly decreases upon being introduced to a task beyond the scope of what it was trained on. Early mitigation strategies focused on enforcing orthogonal input representations [9], establishing that minimising overlapping neuron activation is a generally effective principle, giving rise to the logic behind more current regularisation approaches. Recent approaches start off with dropout [15], which addresses this through temporary noise injection during training, thereby encouraging redundancy in learned representations. This was improved upon with the introduction of fast [20] and adaptive [2] dropout models. However, this remains less effective in RNNs, necessitating the development of zoneout [10], which "randomly replaces some units' activations with their activations from the previous timestep." This was improved upon with surprise-driven zoneout. [1]. While effective, these methods impose no permanent structural change and offer no mechanism for actively renewing neural capacity. More pertinent to our project, the common mechanism behind these techniques is random selection, or notably possessing no strategic mechanism for selecting the most optimal neurons to turnover.

These limitations motivate the investigation of Complementary Turnover Training (CTT), a biologically-inspired paradigm in which neurons are permanently killed and replaced during training. CTT operationalises two well-evidenced biological phenomena: adult neurogenesis in the hippocampal dentate gyrus, where new neurons are selectively born and integrated back into the network [18], and programmed cell death (apoptosis), regulated by homeostatic mechanisms [19]. This maps onto Complementary Learning Systems (CLS) theory [12], which distinguishes a fast-learning hippocampal system from a slow-learning neocortical system, a principle artificial networks have largely ignored. Prior neurogenesis-inspired ML work has demonstrated the viability of turnover-based training. Tran et al. [18] showed neurogenesis-like turnover acts as a neural regulariser on classification tasks, though at the cost of lesion robustness. This is a gap CTT directly addresses. Draelos et al. [6] demonstrated that adding new neurons supports continual learning, and Gurbuz et al. [7] extended this to class-incremental learning without replay. However, these studies treat turnover as a binary intervention and evaluate it exclusively on classification benchmarks. The individual stages of the turnover cycle, specifically the kill/birth ratio and selective neuron scoring mechanism, have not been isolated as experimental variables.

2 Related Work

Here, we discuss the most necessary work in relation to our research objectives and experiments.

2.1 RNN Scale Generalisation in Navigation Tasks

Although studies have been done in tangential topics surrounding maze navigation and a neural network's ability to extrapolate its task functionality to larger sizes, their main focus lies in the features of the maze that result in better or worse performance when extrapolating, rather than the makeup and architecture of the neural networks. It was found that an RNN can successfully extrapolate to larger maze sizes but are less successful under certain maze conditions [3]. Studies further show that RNNs, above all, are best suited to extrapolation compared to other neural network architectures, mainly due to their incorporation of "memory", which allows them to take information from prior inputs to influence current input and outputs [4]. Furthermore, there has been research in other biologically-inspired methods of neural networks applied to maze traversal problems, namely a two-level hierarchical artificial neural network, where the lower level network finds all potential paths, and of the potential paths, the higher-level network finds those which are also valid [14].

It is clear that efforts have been made in the usage of various neural network architectures in maze traversal problems, as well as in the generalisability of these architectures. There has already been an intersection of these two concepts, but with differing research questions. The prior work discussed is a strong indicator that these mediums will serve as good ways to extract meaningful results from our experiments, and address gaps in previous literature that will allow us to further the field of biologically-inspired neural networks and their potential applications.

2.2 Hebbian Learning in Neural Networks

Hebbian learning is a general psychological paradigm that is widely adopted as the general model for long-term memory in humans [11]. More abstractly, it describes a specific relationship in the brain wherein we can imagine an input mapped to a specific output is formed specifically through the strengthening of synaptic connections. [16]. It's of great interest to neuroscientists to represent proposed biological models computationally, given the control one has over the simulation and variables being tested. Further, in trying to replicate the computational efficiency of the human brain, there may be some notable improvement to machine computational efficiency. Thus, Hebbian learning as a paradigm has been implemented as a general learning rule within deep neural

networks [5, 8]. This has been used as a loss function [5] and as its own learning rule in replacement of backpropagation [8]. This is generally considered apt for more biologically sound representation within neural networks [8], as well as some resistance to noisy inputs while maintaining accuracy [5].

3 Problem Statement

The primary aim of this project is to investigate whether CTT produces recurrent neural networks that are more robust, more adaptable, and less prone to catastrophic forgetting than networks trained under static regularisation methods. This investigation is conducted within the context of computational modeling, which provides a controlled and reproducible platform for studying structural plasticity in neural systems, where variables such as turnover, scoring mechanism, and environment complexity can be isolated in ways that biological experimentation alone cannot easily achieve. This hypothesis is motivated by a fundamental limitation of current approaches. Dropout, zoneout, and related methods impose no permanent structural change on the network, and offer no mechanism for the renewal of neural capacity. CTT addresses this directly through a controlled cycle of neuron death and birth, localised to a plastic module and stabilised through homeostasis and consolidation, operationalising principles drawn from adult neurogenesis and Complementary Learning Systems theory.

3.1 Research Objectives

The research objectives and questions addressed by this project are summarised below. Although we share the broader research topic, being in deciding whether CTT is a viable alternative to current memory-based neural network architectures, our research questions differ in isolation of stages within the turnover cycle which are of most interest, namely scoring, killing, and birth. Our two main research questions are as follows:

(1) Does the ratio of kills/births of neurons impact the scale generalisability of a CTT-based RNN while performing maze navigation tasks?

This research question was chosen firstly on the basis of testability, as well as perceived correlation between the amount of neurons killed and replaced. Previous work has been done on turnover-focused methods of regularisation, but with a large focus on small datasets [13]. We hypothesize that balanced ratios will perform better than unbalanced ones due to the stronger stabilization of the network from keeping the same amount of neurons throughout turnover cycles. The reasoning behind the choice of a maze-navigation-centered task is that it can make use of a two-layered biologically-inspired structure where one layer is "short-term" and the other is "long-term". Maze navigation tasks make better use of memory, since agents must keep spatial and sequential information over several time-steps in order to successfully learn the environment. Furthermore, LSTM's (Long Short-Term Memory RNNs) are often used in similar experiments [21], and have been shown to outperform simpler recurrent and feedforward neural networks, but further investigation of whether an LSTM with two mmemory systems will improve navigation performance has not yet been done. Focusing on the steps of the turnover cycle, experimenting with kill-and-birth ratios seems most likely to yield noticeable differences, with the added benefit of having arguably the most control over variables to test, making it easier to experiment with a range of values for more holistic results.

(2) Does Hebbian-scored neural turnover during simulated sleep cycles (where a sleep cycle is considered downtime during which we calculate Hebbian-scores for all neurons) enable more robust long-term and short-term memory consolidation in an RNN foraging agent in comparison to scoring neurons in isolation through CTT?

This research question came about as a potential alternative to the CTT-based scoring methodology, which has been designed and tested as of recent [1]. This is borne from the risk that CTT-based scoring may be subject to overvaluing some neurons that do not necessarily affect the final outcome. Using principles of Hebbian learning, we propose designing an algorithm that instead scores neurons in terms of how largely it influences the overall decision outcome of the neural network downstream, rather than score neurons in isolation. In essence, this is a measurement of the correlation between neurons and their firing per timestep. Further, the reason why sleep and wake cycles are particularly focused on is to mimic and test the Synaptic Homeostasis Hypothesis (SHY), wherein a suspected pruning mechanism in the brain occurs during sleep, and is dependent on information learned during waking periods [17]. Thus, this question further exists to provide a computational basis for testing if Hebbian-based scoring is a likely mechanism for pruning in the brain.

4 Methods

Discussed below is the CTT implementation, which pertains to all project members. Thereafter, it is specifically the methods in relation to the research objectives.

4.1 The Complementary Turnover Training Process

CTT is a biologically-inspired training paradigm in which neurons are permanently removed and replaced during training in a controlled, structured manner. Unlike dropout, which temporarily silences neurons for a single mini-batch before restoring them, CTT imposes persistent structural change, whereby neurons that are killed do not return, and new neurons are introduced to rebuild capacity. The central thesis is that neural systems become more robust, more adaptable, and less prone to catastrophic forgetting through controlled neuronal turnover, localised to a plastic module and stabilised through homeostasis and consolidation.

The turnover cycle consists of five sequential steps executed every K training steps. Each neuron is assigned a combined health score, where importance is the product of the EMA of activation and gradient magnitudes, and redundancy is the maximum pairwise correlation with other neurons. The top 10% of neurons by importance are immune from death, and a minimum living fraction of 70% is enforced. Neurons in the bottom $p\%$ are permanently removed via an alive mask. Each killed neuron is replaced using a clone-and-mutate strategy, copying a high-importance survivor's

[1] https://anonymous.4open.science/r/targeted-neuron-turnover/README.md

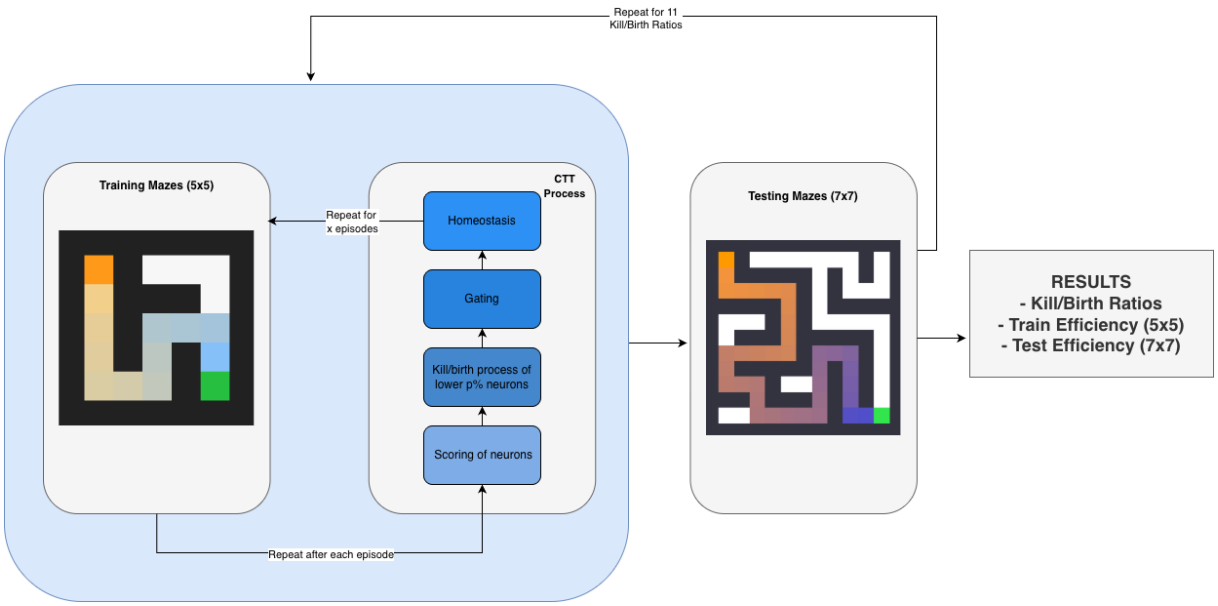


Figure 1: Kill/Birth Experiment Process

weights and adding small Gaussian noise. Newborns are introduced silently at gate = 0 and linearly ramped to full contribution over $T_{\text{gate}} = 2{,}000$ steps, preventing destabilisation of existing representations. Finally, homeostatic scaling and a 2,000-sample replay buffer are applied after each turnover event to protect previously acquired knowledge. It is worth noting within our description of CTT that this does not manifest as a different architecture of the RNN, but instead functions as a regularisation mechanism on top of existing architecture.

4.2 Kill/Birth Ratio Study

Specific to testing research objective 1, we aim to investigate whether the kill/birth ratio within the CTT turnover cycle is a meaningful determinant of scale generalisation performance in an RNN trained on maze navigation. All other CTT parameters are held fixed so that observed differences can be attributed specifically to the ratio. Firstly, we will train baseline neural networks, namely a standard LSTM, an LSTM using dropout, and an LSTM using zoneout on "small" mazes, followed by testing on a batch of unseen slightly larger mazes to provide us with baseline metrics for generalisability performance. Following this, we will implement an LSTM which incorporates CTT and train various combinations of kill/birth ratios to see if we can find a correlation between generalisability performance and specific kill/birth ratios in CTT, as well as how CTT as a whole performs against other regularisation techniques for this particular task. The process is outlined in detail in Figure 1. The maze environment will be fully-observable by the LSTM so that when generalising to larger maze sizes, performance is correlated to how well the model is able to learn a policy for general maze navigation, rather than a specific path or size. Primary metrics are mean time-steps to solve on train-size mazes, mean time-steps to solve on test-size mazes, and the generalisation ratio (test steps / train steps). Secondary metrics include proportion of episodes solved within budget, and living neuron fraction over training.

4.3 Hebbian-based scoring

Using the same baseline neural networks above, alongside the use of a common CTT-based scored LSTM, this experiment will specifically focus on developing an alternative model with Hebbian-based scoring, which will be used to test research objective 2. This is conducted in a fixed-step process. Firstly, a reinforcement-learning based environment will be designed to stress-test an agent's use of long-term and short-term memory. The simulation will have multiple sleep-wake cycles over an environment whose conditions change depending on the current season (as it cycles through summer, autumn, winter, and spring). During waking periods, which will be a fixed number of episodes, the agent will experience the environment, and have its weights and biases updated after every episode. During this, a running mean of these changes per neuron will be tracked. During sleep cycles, rather than calculating scores based off of the CTT-based paradigm, Hebbian-based scoring will instead measure the correlation between the firing of neurons within the network, checking to see whether or not the neuron is actively participating in recurrent dynamics such that it meaningfully impacts the outcome over time. This is to avoid potential over-reliance on neurons with high importance based on gradient impact (CTT), which, if preserved, might still cause the agent to overrely on prior information, potentially forgetting more generalised, correlated neurons that have higher losses. This could result in the agent still suffering from catastrophic forgetting between seasons. During sleep cycles, the correlation matrix will be used as the selection criteria for zeroing out neurons. After which, the kill-birth-gate-homeostasis steps of our CTT paradigm will be

conducted. This repeats until the simulation ends. The key metric for evaluating this will be tracked in two ways: the number of steps the agent survives per episode to evaluate short-term memory consolidation, as well as the number of total episodes the agent survives per season to test effects against catastrophic memory and long-term memory consolidation.

5 Experiments

Our experiments below outline the specific conditions through which we execute the methodologies discussed above to answer our research objectives.

5.1 Experiment 1 - Kill/Birth Ratios

5.1.1 Task and Environment. The RNN must navigate from a fixed start to a goal in a procedurally generated maze. Training uses a number of mazes with separate validation mazes at 5×5 cells (11×11 grid); withheld test mazes at 7×7 cells (15×15 grid) are used exclusively for scale generalisation evaluation. At each time-step the network receives the full binary grid, a one-hot agent position, and normalised goal coordinates. A step penalty of −0.01, wall penalty of −0.1, and goal reward of +10.0 constitute the reward signal, with training via REINFORCE and gradient clipping. The experiment process can be seen clearly in Figure 1.

5.1.2 Baseline conditions. A vanilla LSTM and dropout LSTMs at p=0.1,0.2,0.5; p = 0.1, 0.2, 0.5; p=0.1,0.2,0.5 provide the primary comparison. A zoneout baseline is included as an LSTM-specific reference. A pilot study varying hidden size across 64, 128, and 256 units and learning rate across 10^{-4} to 10^{-2} is conducted before the full sweep to confirm the architecture is in an appropriate complexity regime.

5.1.3 CTT Conditions. Eleven CTT conditions span three regimes (shrinking, balanced, and net growth) plus a death-only ablation to isolate the birth step's contribution. Balanced 1:1 conditions are the primary hypothesis variants. Each condition is run across a minimum of three seeds and reported as mean ± std. The full set of conditions is summarised in Table 1.

Table 1: CTT Kill/Birth Ratio Conditions

Condition	Kill	Birth	Ratio	Regime
Death-only	5%	0%	∞:0	No birth
Unbalanced (low)	5%	2%	2.5:1	Shrinking
Balanced (low)	5%	5%	1:1	Balanced
Net growth (low)	5%	10%	1:2	Growing
Unbalanced (mid)	10%	5%	2:1	Shrinking
Balanced (mid)	10%	10%	1:1	Balanced
Net growth (mid)	10%	20%	1:2	Growing
Unbalanced (high)	20%	10%	2:1	Shrinking
Balanced (high)	20%	20%	1:1	Balanced
Net growth (high)	20%	30%	1:1.5	Growing
High balanced	30%	30%	1:1	Balanced

If we find that there are specific turnover ratios which perform noticeably better than others, then we can conclude that this stage of the turnover cycle does play a role in the scale generalisability of maze navigation tasks.

5.2 Experiment 2 - Hebbian-Based Scoring

To test both long term and short term memory, this experiment relies on a carefully designed environment. We will implement an LSTM in an episodic-learning environment, borrowing from reinforcement learning principles. Thus, the environment must be designed in such a way that:

(1) The agent must learn something during its waking period (e.g. which patches are good relative to the season)
(2) The agent must remember it across time (e.g. it must not forget spring knowledge during summer)
(3) The agent must consolidate general rules (e.g. which patch is best given its energy).

The environment will have a few key phases:

(1) The agent will have a certain number of steps per episode to forage
(2) The agent will experience all four seasons
(3) Each season will have a certain number of episodes. Backpropagation will be done per episode. After which, the agent will undergo "sleep" in which the Hebbian scores are calculated and neurons are replaced.
(4) There will be a certain number of wake-sleep cycles per season to ensure short-term memory consolidation alongside long-term memory task functions.

The exact steps of the environment is outlined in Figure 2 below:



Figure 2: Foraging agent learning process

In our environment, a season can either be labelled as "harsh" or "mild", which will be split an even number of times over the

number of cycles per season. This dictates the energy level the agent has, wherein if the season is mild, the agent has more energy. Its energy will be directly proportional to the number of steps per episode, and baselined for mild seasons. We want this so that in mild episodes, an agent dies if it does not encounter a single patch of food once throughout the episode at all. In harsh episodes, the agent is more conditioned to find optimal paths and better food patches to ensure survival. To comparatively gain energy, the agent will have to forage food from patches strewn across the environment randomly per full simulation, i.e. for the entire simulation, the patches will be randomly initialised, then continue to remain in the same spot. Patches within the same season will be placed relatively close to each other. If the patch is in-season, it gives the agent the most energy. If it's in an adjacent season (e.g. the adjacent seasons of summer are autumn and spring), it replenishes a little bit more energy than it would if it were out of season, but still less than it would if it were in season. There will also be different yields per patch, where higher yield patches replenish more energy, since you consume more at a given step. Thus, the agent here must create strong associations with harshness and seasonal cycles to optimise the best path for survival. We add more noise and conditions to the environment in three ways:

(1) A regrowth period, in which patches are able to regrow within an episode, where patches in-season grow comparatively fastest, and patches out of season grow comparatively slowest.
(2) Toxicity, in which one patch per season is toxic to eat, and will kill the agent if it does. Similarly, the probability of a patch is dependent on its seasonality.
(3) Environmental obstacles. The environment will have some obstructions that will negatively reward the agent if it runs into them. Naturally, the environment borders are an obstacle. However, rocks/walls will be implemented randomly around the environment per simulation, such that the agent will also have to learn optimal paths to patches with some level of obstruction.

Note that the agent will not be made aware of the entire environment. This is so that it doesn't rely on any global observation information, and must inform its decisions based on its local observations at that given moment. If it is found that the agent's short-term and long-term survivability noticeably differs over time, in comparison to the baseline and the CTT-based scored models, then it will have answered the research objective of the experiment.

Please see Table 2 in Appendix A for a full summary of experiments.

6 Ethical, Professional and Legal Issues

No foreseeable legal issues as environments are designed by ourselves. Experiment 1 is using publicly available and open source code to generate mazes. Experiment 2 is entirely self-designed given how particular the conditions are. We have no need for human involvement in our experiment and the foreseeable outcomes of the experiment don't pose any ethical issues. Furthermore, we will adhere to the Open Source Software guidelines and as such will use our neural networks in a professional and ethical manner.

7 Contributions

This project makes two contributions to the study of biologically-inspired recurrent neural networks. First, by systematically varying the kill/birth ratio across 11 CTT conditions evaluated on a maze scale generalisation task, we provide the first controlled study of this parameter within the CTT framework, producing empirical evidence for or against the hypothesis that balanced turnover produces more robust representations than unbalanced or death-only regimes. Second, by replacing CTT's scoring mechanism with a Hebbian-based alternative applied during simulated sleep cycles, we provide a computational evaluation of the Synaptic Homeostasis Hypothesis in a foraging agent, establishing whether biologically-motivated scoring produces measurably better memory consolidation than gradient-based scoring. Together, these contributions advance the understanding of which components of the CTT cycle drive its performance benefits, and provide a foundation for future work on biologically-grounded continual learning in RNNs.

8 Project Plan

Below are specific points regarding the pragmatic outline of performing our project throughout the year.

8.1 Risks

The project risks are given in the risk matrix in Appendix B.

8.2 Timeline

The project began on the 25th of March 2026 and concludes with final code and website submission on the 14th of September 2026. The full scheduling and division of work, including task dependencies and milestone dates, are detailed in the Gantt chart found in Appendix B. Key phases include experimental setup and baseline implementation (April–May), CTT and Hebbian scoring implementation (May–June), full experiment sweep (July), analysis and write-up (August–September).

8.3 Resources Required

- UCT High Performance Cluster
- Open-source CTT implementation: https://anonymous.4open.science/r/targeted-neuron-turnover/README.md
- Personal access to a computer and appropriate IDE

8.4 Deliverables and Milestones

The main deliverable for this project is the final research paper. Key deliverables and their target dates are detailed in the Gantt chart (Appendix B); notable milestones include:

- Two literature reviews — early April 2026
- Project proposal — early May 2026
- Software feasibility demonstration — early July 2026
- Successful CTT implementation across both experiment tracks — end June 2026
- Hebbian scoring implemented for Experiment 2 — mid July 2026
- Project report draft — mid August 2026
- Final project report — mid September 2026
- Final code submission — mid September 2026

- Project website and poster — October 2026

8.5 Work Allocation

The work of coding the various RNN models, including a base model, a model with dropout, zoneout, and CTT-based scoring will be developed by both students. Nathan Del Fava will be implementing experiment track 1, and is responsible for developing the maze training data. Ziyaad Shaboodien will be implementing experiment track 2, and is responsible for designing the environment and implementing Hebbian-scoring.

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A Summary of Experiments

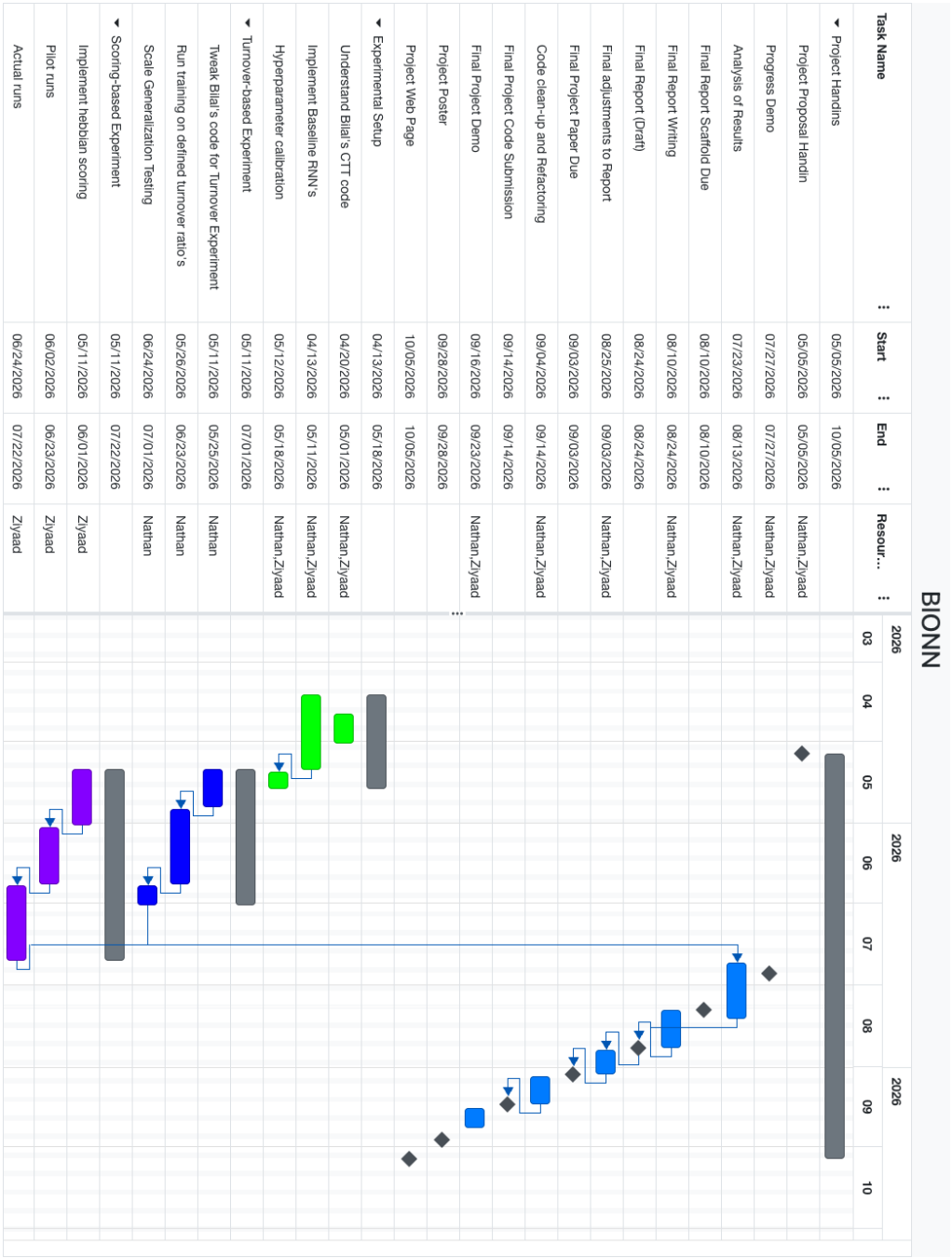
Table 2: Summary of Experiments

	Experiment 1: Kill/Birth Ratio	**Experiment 2: Hebbian-Based Scoring**
Baselines	Vanilla LSTM, LSTM w/ Dropout, LSTM w/ Zoneout	Vanilla LSTM, LSTM w/ Dropout, LSTM w/ Zoneout
Research Question	Does the ratio of kills/births of neurons impact the scale generalisability of a CTT-based RNN while performing maze navigation tasks?	Does Hebbian-scored neural turnover during simulated sleep cycles (where a sleep cycle is considered downtime during which we calculate Hebbian-scores for all neurons) enable more robust long-term and short-term memory consolidation in an LSTM foraging agent in comparison to scoring neurons in isolation through CTT?
Environment	Procedurally generated mazes; Smaller training maze, scaled up for testing; Reinforce with step/wall/goal reward signal.	Episodic foraging environment; Summer, Autumn, Winter, Spring; Patches of food with varying yields, corresponds to different seasons; Toxicity, regrowth periods, and environmental obstacles; Sleep-Wake cycles mimicking Night/Day.
Testing	11 CTT kill/birth ratio conditions spanning shrinking, balanced, and net-growth regimes (see Table 1); Compared to baselines	Hebbian-scored model v CTT-scored model v baselines
Key Metrics	Steps per maze (small); Steps per maze (large); Kill/Birth ratios	In-season / per-season survivability

B Risk Matrix

Risk	Prob.	Impact	Strategy
CTT implementation diverges between experiment tracks	Low	High	Develop CTT logic together into a central module, imported into both tracks then separate modules
Gated newborn neurons cause gradient instability, causing training to diverge after one turnover	Low–Med	High	Monitor loss curve at every respective turnover boundary. If outliers or spikes, increase the gating value and consider gradient clipping mechanisms
Local computer architecture is too weak to conduct plasticity experiments	Med	Med	Given HPCs are available, track experiments alongside HPCs if needed
Inability to find meaningful relationships in experiment results and the hypothesis	Med	Med	Research alternative scoring / scalability mechanisms to further test our research objectives
HPC job queue delays push timeline past submission window	Med	Med	Ensure majority of tests are done before requiring HPCs

C Gantt Chart



Task Name	Start	End	Resour...
▾ Project Handins	05/05/2026	10/05/2026	
Project Proposal Handin	05/05/2026	05/05/2026	Nathan,Ziyaad
Progress Demo	07/27/2026	07/27/2026	Nathan,Ziyaad
Analysis of Results	07/23/2026	08/13/2026	Nathan,Ziyaad
Final Report Scaffold Due	08/10/2026	08/10/2026	
Final Report Writting	08/10/2026	08/24/2026	Nathan,Ziyaad
Final Report (Draft)	08/24/2026	08/24/2026	
Final adjustments to Report	08/25/2026	09/03/2026	Nathan,Ziyaad
Final Project Paper Due	09/03/2026	09/03/2026	
Code clean-up and Refactoring	09/04/2026	09/14/2026	Nathan,Ziyaad
Final Project Code Submission	09/14/2026	09/14/2026	
Final Project Demo	09/16/2026	09/23/2026	Nathan,Ziyaad
Project Poster	09/28/2026	09/28/2026	
Project Web Page	10/05/2026	10/05/2026	
▾ Experimental Setup	04/13/2026	05/18/2026	
Understand Bilal's CTT code	04/20/2026	05/01/2026	Nathan,Ziyaad
Implement Baseline RNN's	04/13/2026	05/11/2026	Nathan,Ziyaad
Hyperparameter calibration	05/12/2026	05/18/2026	Nathan,Ziyaad
▾ Turnover-based Experiment	05/11/2026	07/01/2026	
Tweak Bilal's code for Turnover Experiment	05/11/2026	05/25/2026	Nathan
Run training on defined turnover ratio's	05/26/2026	06/23/2026	Nathan
Scale Generalization Testing	06/24/2026	07/01/2026	Nathan
▾ Scoring-based Experiment	05/11/2026	07/22/2026	
Implement hebbian scoring	05/11/2026	06/01/2026	Ziyaad
Pilot runs	06/02/2026	06/23/2026	Ziyaad
Actual runs	06/24/2026	07/22/2026	Ziyaad